

Research and application of business intelligence and data mining based on big data

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Abstract. In the environment of increasingly fierce market competition, more and more enterprise managers realize the importance of business intelligence and data mining to enterprise management. It has become the consensus of many manufacturing enterprises to mine potential decision information from these data by means of informatization, support enterprises to complete strategy formulation and help enterprises process data more efficiently. This paper discusses the theoretical development, research status and application status of business intelligence and sales management, in order to provide a theoretical basis for this research. This paper proposes to use business intelligence technology to establish a data analysis platform. Through data warehouse technology, online processing analysis technology and data mining technology, it helps enterprises complete data collection, analysis, and mining in the process of business intelligence management, improves the accuracy of business intelligence data processing, and reduces the manpower and time consumption of business intelligence data processing, it makes up for the limitations of the existing sales information management system. Finally, data mining based on business intelligence can improve the company's management decision-making and execution ability, and enable large enterprises to mine data and effective information more intelligently and accurately.

Keywords-business intelligence, data mining, big data

I. Introduction

Big data is the massive data generated in the era of information explosion, while business intelligence is a word that only appeared in recent years. It began to become an upsurge in 2012 and has almost become a household name so far. The typical characteristics of big data can be summarized into four V's, namely, large amount of data, various types, low value density and fast speed. There are many fields of big data application, covering all corners of life. Business intelligence [1-3] refers to the use of modern data warehouse technology, online analysis and processing technology, data mining and data presentation technology for data analysis to realize business value. For enterprises, it is a common data support means. It is a complete solution to effectively integrate the existing data in the enterprise, quickly and accurately provide reports and put forward decision-making basis, and help enterprises make wise business decisions. Business intelligence is generally understood as a tool to transform existing data into knowledge and help enterprises make informed business decisions. With the continuous reform of China's enterprise management system, especially under the background of the rapid development of information technology, business intelligence has been widely used in enterprise management. Especially for some large and medium-sized enterprises, it has

achieved good results in the application of business intelligence. However, in the era of big data [4-7], some enterprises still have many weak links in building personalized business intelligence, especially in many aspects such as data mining, integration, analysis, processing and application using big data technology, which need to be highly valued by enterprises. Big data has the characteristics of strong openness, integration and sharing, which can provide strong support for the construction of personalized business intelligence. As shown in Figure 1, business intelligence is a data factory.

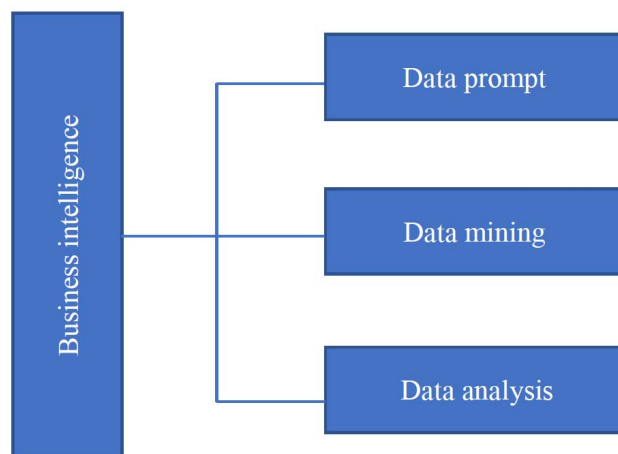


Figure 1. Business intelligence is a data processing factory.

II. Important value of big data to personalized business intelligence

The important value of big data to business intelligence is that it can realize the collection, processing and analysis of a large number of business intelligence data, so as to change the paradigm of traditional business intelligence data application. By constructing relevant models in the field of business intelligence, it explores the relevant relationship between business intelligence variables, and provides effective support for business intelligence teaching and decision-making, so as to realize the personalization and Diversified evaluation and scientific decision-making.

III. Constraints on building personalized business intelligence in the era of big data

A. lack of innovation in application platform

For personalized business intelligence, in order to achieve real results in the application of big data technology, we must

make great efforts in building the carrier of personalized business intelligence big data platform. Only in this way can we create favourable conditions for more effective application of big data technology. From the current overall situation of some enterprises building personalized business intelligence, although it can be effectively combined with the actual situation of enterprises, there are still many weak links in the application of big data technology, especially in the construction of personalized business intelligence big data platform.

B.the application system is not perfect

Many enterprises do not pay attention to improving and perfecting the big data application system, resulting in imperfect big data application system and insufficient internal and external resource mining, integration and application, which will inevitably restrict the effectiveness of personalized business intelligence. Some enterprises do not pay attention to the systematic application of big data technology, especially the collection of various data resources is not in place, and even many departments do not pay attention to the authenticity, accuracy and continuity of data, and even there is data fraud, which directly leads to the limitation of the application of business intelligence. Even for this reason, many enterprises give up business intelligence.

IV.Data mining algorithm

The system uses a variety of data mining algorithms to improve the efficiency and accuracy of business intelligence.

A.data algorithm based on association rules

Recommendation based on association rules is to count the relationship between different rules according to historical data, such as $X \rightarrow Y$, which means that after X event occurs, y event will have a certain probability. This probability is obtained through historical data statistics. The purpose of association rules is to find the relationship between items in a data set, also known as market basket analysis. For example, 10% of the customers who buy shoes may also buy socks, and 60% of the customers who buy bread will also buy milk. The most famous example of this is the story of "diapers and beer". For a rule $X \rightarrow Y$, there are two indicators to measure the rule. One is support, which indicates the proportion of all sample data including both X and Y samples. The other is confidence, which indicates the proportion of samples containing y in all samples containing X.

B.deep learning algorithm

Deep learning (hereinafter referred to as DL), in other words, can be said to be machine learning based on artificial neural network. Different from traditional machine learning, DL needs more samples in exchange for less manual annotation and higher accuracy. Convolutional neural network is a kind of artificial neural network, which has become a research hotspot in the field of speech analysis and image recognition. Its weight sharing network structure makes it more similar to biological neural network, which reduces the complexity of network model and the number of weights. This advantage is more obvious when the input of the network is a multi-dimensional image, so that the image can be directly used as the input of the network, avoiding the complex process of feature extraction and data reconstruction in the traditional recognition algorithm.

C.text mining

Here are the six steps of text mining: get text; reprocessing text; linguistic processing of text stream; mathematical processing of text stream; feature extraction and feature selection and mining using algorithms

D.web mining process

The general process of Web Data Mining: Web data resource acquisition; Web data reprocessing; data conversion and integration; pattern recognition and pattern analysis.

V.System architecture

Driven by the national big data strategy, big data technology has developed rapidly. In enterprise business intelligence decision-making, big data presents great enterprise innovation value. According to the needs of enterprise data analysis, a big data business intelligence analysis platform based on distributed storage architecture is designed and implemented. The big data business intelligence platform based on distributed storage architecture is composed of data access, data storage, data processing and business intelligence applications. The data source of the data access layer is multi-source heterogeneous data, which is collected manually and by equipment, mainly including SAP, non-sap, XML and other data sources. The system architecture of business intelligence system is shown in Figure 2.

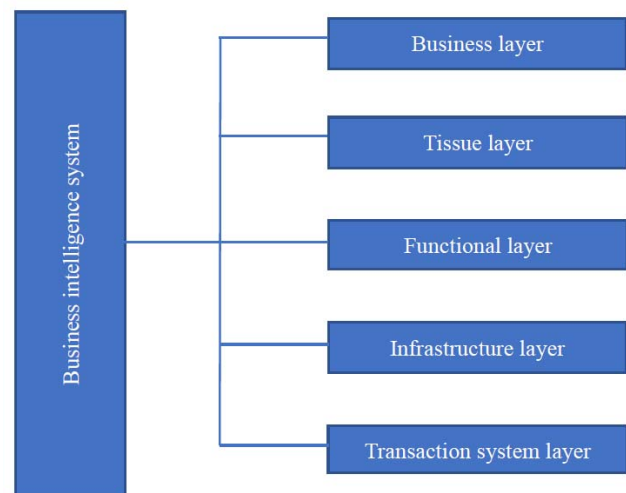


Figure 2. Business intelligence system architecture diagram.

Ensuring data integrity and consistency is an important basis for data access and storage. The amount of data faced by enterprises has jumped from TB to Pb, and the growth rate of data storage capacity lags far behind that of data. Designing a reasonable distributed data storage architecture has become a key challenge. The data storage layer of the system adopts distributed storage technology to store the original data of the data access layer in the distributed file system HDFS, HBase and traditional databases (such as Oracle, SQL server, MySQL.), and access and process the data through the distributed storage interface of Hadoop / spark big data platform.

The system combines intelligent processing with storage. Through integrated analysis, enterprises can not only make full use of the existing storage infrastructure, but also carry out effective resource monitoring. Based on Hadoop / spark big data platform, the data processing layer calculates, analyses and processes the data of the data storage layer through the distributed / parallel computing programming interface, such as data selection, data cleaning, data integration, data mining [8-10] and model operation, so as to provide distributed computing support for the application layer. Through the data mining and pattern recognition of the lower data, the business intelligence application layer provides decision-making applications such as enterprise management analysis, business comprehensive analysis, support decision-making analysis, operation monitoring analysis and performance appraisal analysis according to the specific application needs.

VI. Big data intelligent financial statement design

Intelligent financial statement design is an important basis for business intelligence analysis and application. The intelligent accounting report design based on big data mainly includes two parts: data model design and financial report optimization design. It can be divided into main data layer design, data extraction layer design, cleaning and conversion layer design, report analysis layer design and report query optimization design. Figure 3 shows the Intelligent financial statement design.

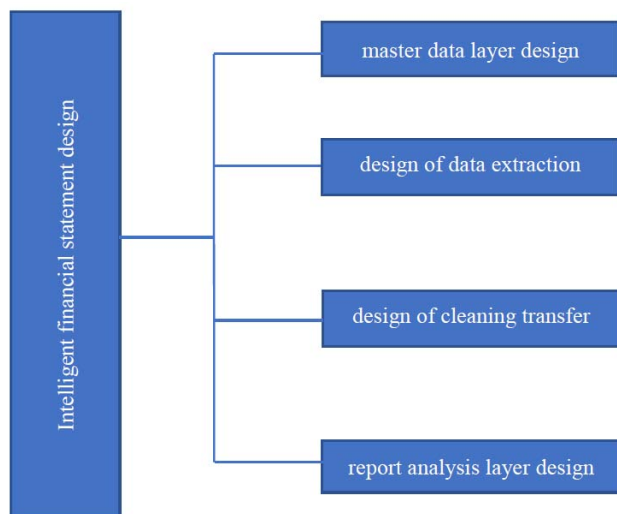


Figure 3. Intelligent financial statement design diagram.

A. master data layer design

The master data layer stores information objects, including standard information objects and user-defined information objects. Establish conversion relationship between information object and data source.

B. design of data extraction layer

The data extraction layer is the first layer model built. The stored data comes from the most original data of the system. The main goal is to design and write optimized data extraction logic. The directly obtained data source is stored in the data

storage logic. The data loading method is mainly to extract the relevant field information by performing incremental loading. The data extraction layer adopts the cumulative data storage method to distinguish the data through technical keywords (request number, data package number and data record number).

C. design of cleaning transfer layer

The clean transformation layer is the core model for analysing and processing write optimization data. All logical transformation relationships are implemented here. The model mainly adopts standard data extraction logic and is composed of three tables: new table, activity table and log table.

D. report analysis layer design

The model of report analysis layer is the top-level analysis model of business information warehouse and data warehouse. This layer carries the analysed data from the cleaning conversion layer, and maintains the consistency with the standard data extraction logical model data. However, the model at this level is mainly information cube, some standard data extraction logic, and the final multi-information provider. The information cube has the same cumulative data characteristics as the write optimization data extraction logic, and the key values and dimension levels to be analysed are simpler.

VII. Conclusions

With the continuous development of China's economy and the continuous enrichment of various data resources, the establishment of business intelligence system has become a development trend of large and medium-sized enterprises in China. However, it is difficult to establish business intelligence system because of its unclear initial demand and large investment scale. China's business intelligence system is still in the introduction period, and there is still a lot of work to be done on the research of business intelligence. In order to achieve business growth, it is undoubtedly very important for enterprises to pay attention to the current development trend of business intelligence system and the relationship between these trends and enterprise decision-making process.

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